Implementation of DeFT: A Deadlock-Free and Fault-Tolerant Routing Algorithm for 2.5D Chiplet Networks

Yusuf Tahir Orhan - N24112049*
*Computer Engineering Department, Hacettepe University, Ankara, Turkey
†Graduate School of Science and Engineering, Hacettepe University, Ankara, Turkey
E-mail: *yusuftahirorhan@hacettepe.edu.tr

***Abstract*—This project implements and evaluates a Noxim-based `DEFT_2_5D` simulator extension for studying DeFT-style routing in a 2.5D chiplet network. The implementation models four chiplets on an active interposer, a deterministic physical Vertical Link (VL) inventory, startup permanent VL faults, DeFT virtual-network assignment rules, movement-transition filtering, schema-v1 offline VL lookup tables, explicit `XY` baseline modes, synthetic traffic profiles, and machine-readable metrics export. A fixed-window synthetic-traffic sweep completed 150 simulator runs across routing modes, traffic profiles, fault masks, and seeds. Those measurements are reported descriptively because the finite simulation window contains empty and partial injection cells, including 54 completed runs with zero packets injected after the warm-up boundary. A separate opt-in drain-mode DeFT reachability audit provides stronger evidence within its own boundary: under deterministic hardcoded fixtures, seed 0, the current physical bidirectional VL fault model, and accepted masks `0x0000`, `0x0001`, `0x0011`, `0x0111`, and `0x1111`, all 650 cases passed and all 4032 valid ordered chiplet-router source-destination pairs were delivered for every accepted mask. The report therefore separates finite-window traffic behavior from eventual-delivery validation after injection cutoff and drain, and avoids unsupported latency, throughput, ranking, statistical, PARSEC/GEM5, or directional-fault claims.**

***Index Terms*—2.5D integration, chiplet networks, Network-on-Chip, deadlock-free routing, fault tolerance, Noxim.**

I. Introduction

2.5D chiplet integration connects multiple smaller dies through an active interposer. This architecture can improve modularity and yield, but it also makes inter-chiplet routing sensitive to VL availability and deadlock-avoidance constraints. DeFT is a routing approach proposed for this setting; it combines virtual-network assignment rules with fault-aware VL selection so inter-chiplet packets can be routed through the interposer while respecting deadlock-avoidance constraints [1].

The goal of this project is to extend the cycle-accurate Noxim simulator [2] so DeFT-style behavior can be studied in a reproducible workflow. The implementation focuses on a four-chiplet system with 4×4 chiplet-local meshes, a modeled active interposer layer, four physical bidirectional VLs per chiplet, permanent startup VL faults, two virtual channels for VN.0 and VN.1, and descriptive final metrics for reachability, average latency, and network throughput.

The current implementation models 16 physical bidirectional VLs. This physical fault model differs from single-

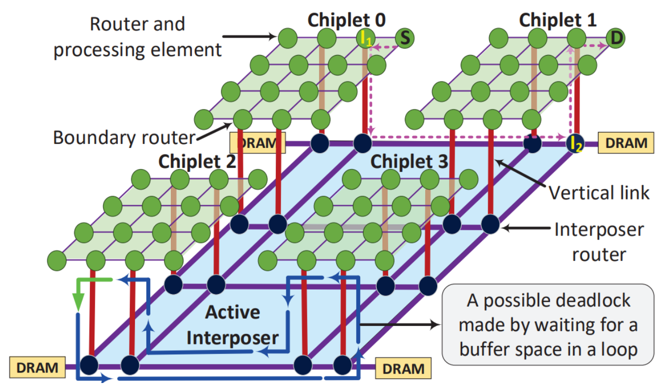

Fig. 1. Baseline 2.5D network architecture reused from the extended proposal source template.

direction endpoint fault modeling in the original paper and is treated as an evaluation constraint throughout the report.

II. Related Work

Routing for chiplet-based and interposer-based systems has been studied through several deadlock-avoidance and reliability mechanisms. Modular turn restriction approaches such as MTR restrict dependency cycles in chiplet systems [3]. Remote-control and update-packet bypassing ideas explore alternate mechanisms for modular systems and 2.5D deadlock handling [4], [5]. ReD extends the virtual-network reliability direction for 2.5D interposer systems [6]. Fault-tolerant 3D NoC routing and configurable inter-die links provide related reliability context, but their assumptions differ from the up-down-up communication pattern of active-interposer 2.5D systems [7]–[9].

This project uses the original DeFT paper as the primary algorithmic reference for virtual-network assignment, movement restrictions, and design-time VL selection [1]. The implementation does not claim a new formal proof; it implements and validates the documented project surfaces inside Noxim.

III. Problem Definition

The project problem is to support deadlock-free and fault-tolerant inter-chiplet routing in a modeled 2.5D system while preserving traceable simulator behavior. The target topology contains four CPU chiplets integrated on an active interposer. Each chiplet contains a 4×4 mesh, and each chiplet has four physical bidirectional VLs located at boundary routers.

The evaluation target from the project requirements is synthetic traffic under permanent VL fault scenarios, using reachability, latency, and throughput metrics. The implemented synthetic profiles are uniform, localized with 40% same-chiplet traffic probability, and hotspot traffic with three deterministic hotspot routers. Real-application PARSEC/GEM5 trace evaluation remains outside the current validated artifact set; PARSEC is a benchmark suite reference point, but no PARSEC/GEM5 trace workflow is validated in this final report [10].

IV. System Architecture and Design

A. Topology Model

The `DEFT_2_5D` topology uses a 2×2 chiplet grid. Chiplet router IDs are `0..63`; interposer router IDs are `64..127`. Packet sources and final destinations remain chiplet routers only. The interposer routers are internal transit routers.

Each chiplet owns four physical bidirectional VLs at deterministic boundary routers. The current model therefore has 16 physical bidirectional VLs. These physical links can also be described as 32 directional-equivalent channels for paper-aligned fault-rate reporting, but the simulator fault model disables one physical bidirectional VL as one object.

B. Virtual Networks

For `DEFT_2_5D`, VN.0 maps to VC 0 and VN.1 maps to VC 1. This avoids a parallel virtual-network field and keeps the routing state aligned with the existing Noxim virtual-channel flow-control path. Boundary-router reassignment is output-VC-aware so a head flit can reserve and forward into the selected VN safely.

The implemented movement filter enforces the current VN transition restrictions: VN.1 to VN.0 is rejected, Up-to-Horizontal movement in VN.0 is rejected, and Horizontal-to-Down movement in VN.1 is rejected. Because explicit Up and Down ports do not exist in the original Noxim direction set, the current implementation derives movement context from the layer, boundary-router state, input direction, and output direction.

C. VL Lookup Tables

The offline LUT generator emits a versioned `deft_vl_lut.v1` YAML format. Runtime DeFT routing loads the schema-v1 LUT, derives the active physical fault mask from current VL state, and selects source-side exit and destination-side entry VL records by lookup. Missing entries or nonfunctional selected VLs fail closed by returning no legal output direction.

V. Implementation

The simulator extension follows Noxim's existing topology, routing, configuration, and statistics surfaces. The main implemented components are:

- `external/noxim/src/DeftTopology.*`: layer-aware router mapping, physical VL records, boundary-router inventory, and topology validation.
- `external/noxim/src/DeftFaultInjectionManager.*`: startup permanent physical VL fault configuration and validation.
- `external/noxim/src/DeftVirtualNetwork.*`: VN/VC mapping helpers and DeFT VN assignment support.
- `external/noxim/src/routingAlgorithms/Routing_DEFT.*`: runtime DeFT route selection through schema-v1 VL LUT entries.
- `external/noxim/src/DeftVerticalLinkLut.*`: schema-v1 VL LUT loader and validator.
- `external/noxim/config_examples/`: `DEFT_2_5D` topology, `XY` baselines, and synthetic traffic configurations.
- `external/noxim/other/deft_vl_lut_generator.py`: deterministic offline VL LUT generator.
- `external/noxim/other/deft_experiment_runner.py`: experiment launcher.
- `external/noxim/other/deft_analysis_artifacts.py`: analysis artifact generator.
- `external/noxim/src/GlobalStats.*` and `external/noxim/src/ProcessingElement.*`: machine-readable metrics export and injected-packet counters.

Generated artifacts under `external/noxim/other/generated/` are used as reproducibility records and remain traceable to their manifests.

VI. Experimental Setup

The final sweep was defined before running the matrix and was executed without changing simulator behavior during report drafting. Table I summarizes the matrix. The fixed-window sweep contains exactly 150 planned and completed simulator runs. It does not include a post-injection drain phase.

A follow-up diagnosis classified the current standard `XY` implementation as a limited control baseline on `DEFT_2_5D`, not as an interposer-aware unrestricted inter-chiplet baseline. The existing `XY` route path selects only cardinal directions from the global footprint. It does not select VL, hub, interposer, or chiplet-layer phases. On `DEFT_2_5D`, unrestricted inter-chiplet routes require VL/hub/interposer traversal, so standard `XY` can select chiplet-layer movement that has no physical cross-chiplet cardinal link.

The later drain-mode comparison therefore uses `INTERPOSER_AWARE_XY`, a separate baseline that can traverse the modeled interposer path. This baseline is not standard `XY`, and this report keeps those labels separate.

A post-injection drain/source-cutoff policy is needed for eventual-delivery analysis. That policy does not by itself fix standard `XY` topology incompatibility on unre-

TABLE I
FINAL SWEEP MATRIX

Dimension	Values
Routing modes	`XY`, `DEFT`
Traffic profiles	`uniform`, `localized_40`, `hotspot_3x10`
Fault masks	`0x0000`, `0x0001`, `0x0011`, `0x0111`, `0x1111`
Physical fault rates	0%, 6.25%, 12.5%, 18.75%, 25%
Seeds	`0`, `1`, `2`, `3`, `4`
Simulation window	`-sim 10000`
Warm-up window	`-warmup 1000`
Stats format	JSON

stricted inter-chiplet traffic, because standard `XY` still has no VL/hub/interposer route phase.

The drain-mode evaluations use opt-in source-cutoff plus drain semantics for deterministic hardcoded fixtures. They are not fixed-window final sweeps and are not substitutes for the synthetic-traffic tables; instead, they answer the separate question of eventual delivery after injection stops and the network drains.

A zero-injection row is a completed simulator run with `total_injected_packets == 0` in the measured stats window. It is not treated as a failed simulator invocation. Warm-up-0 diagnostics are used only as traceability evidence that traffic sources were not empty; they are not replacement performance data. The fixed-window runner policy does not evaluate eventual delivery after a post-injection drain phase, and the physical bidirectional fault model does not represent the original paper's single-direction 3.125% fault case.

VII. Validation Provenance

The fixed-window synthetic-traffic results are traceable through the following generated artifacts:

- T0026 final sweep output: `external/noxim/other/generated/t0026_final_sweep_v1/`.
- T0026 final analysis output: `external/noxim/other/generated/t0026_final_analysis_v1/`.
- T0027 report-support output: `external/noxim/other/generated/t0027_report_support_v1/`.
- T0028 report results draft: `external/noxim/other/generated/t0028_final_report_results_v1/report_results_draft.md`.

The executed fixed-window manifest records `mode: execute`, `run_count: 150`, 150 completed runs, and 150 return code 0 runs. All 150 JSON stats files were present and contained the exported metric fields. The derived condition and pair tables were cross-checked against the raw statistics and generated summaries.

A separate diagnosis explains the source of the `XY` blank cells by inspecting the runner warm-up policy, injected-packet accounting, standard `XY` routing, and `DEFT_2_5D` topology wiring. It does not add new fixed-window performance rows, does not replace the 150-run sweep, and does not change simulator behavior.

Additional drain-mode provenance is traceable through:

- T0060 DeFT reachability audit: `external/noxim/other/generated/t0060_deft_reachability_audit_v1/`.
- T0053 IA-XY versus DeFT drain comparison: `external/noxim/other/generated/t0053_drain_iaxy_deft_comparison_v1/`.

The T0060 audit used `DEFT` only, seed 0, opt-in drain mode, deterministic hardcoded fixtures, a generated schema-v1 LUT, and accepted physical fault masks `0x0000`, `0x0001`, `0x0011`, `0x0111`, and `0x1111`. It recorded 650 cases, 650 pass cases, zero timeout or non-100% cases, and all 4032 valid ordered chiplet-router source-destination pairs delivered for every accepted fault mask. The T0053 comparison matrix is retained only for denominator-safe IA-XY comparison context: all `DEFT` rows passed, while IA-XY passed 68 rows and timed out in 27 rows.

VIII. Results and Evaluation

This section separates two measurement regimes. Tables II–VI report finite-window synthetic-traffic behavior from the 150-run sweep; blank cells indicate that the supporting denominator is absent. Section VIII-A6 reports deterministic drain-mode validation after source cutoff and network drain. The latter validates eventual delivery within its fixture boundary and does not reinterpret the finite-window metrics.

A. Performance Results

This subsection reports measured fixed-window outputs and then the separate drain-mode reachability audit. The fixed-window tables describe the final 150-run synthetic sweep, while the drain-mode table describes deterministic eventual-delivery validation.

All 150 planned simulator runs completed with return code 0, and the generated report-support tables were cross-checked against the executed manifest, raw JSON stats, sweep summary, analysis run summary, and analysis comparison summary with zero mismatches.

The measured data set contains completed runs in which no packets were injected after the warm-up boundary. It records 54 individual zero-injection runs. At condition level, 12 cells have packet injection in all five seeds, 13 cells have packet injection in only some seeds, and 5 cells have no packet injection in any seed. The 5 empty-injection cells are all `XY` with `hotspot_3x10` traffic.

Under this fixed-window measurement policy, the `XY` hotspot cells cannot be compared with `DEFT` because the `XY` side injected zero packets in every hotspot fault-mask cell. Diagnosis showed that these `XY|hotspot_3x10` zero-injection cells are measured-window artifacts: with `-warmup 1000`, early `XY` traffic had already injected before the measured window and then stalled on topology-incompatible paths. Warm-up-0 diagnostic values are kept only as traceability evidence and are not reported as final performance results.

TABLE II
ARTIFACT AND READINESS SUMMARY

Item	Value
Planned and completed runs	150
Raw stats rows	150
Condition cells	30
XY/DEFT pair rows	15
Individual zero-injection runs	54
Complete-injection condition cells	12
Partial-injection condition cells	13
Empty-injection condition cells	5
Cross-check mismatches	0
Interpretation scope	Descriptive fixed-window metrics

TABLE III
COVERAGE BY ROUTING AND TRAFFIC

Routing	Traffic	Runs	Nonempty	Zero	Injected	Received
DEFT	hotspot_3x10	25	22	3	336	134
DEFT	localized_40	25	25	0	1016	542
DEFT	uniform	25	24	1	436	238
XY	hotspot_3x10	25	0	25	0	0
XY	localized_40	25	20	5	35	0
XY	uniform	25	5	20	5	0

The XY uniform and localized cells injected packets in some seeds, but received zero packets in the measured window. Diagnosis classified these XY|uniform and XY|localized_40 zero-received cells as standard XY topology incompatibility on DEFT_2_5D: cardinal-only XY cannot route unrestricted inter-chiplet traffic through the VL/hub/interposer path required by this topology. Therefore, side-by-side latency comparisons against DEFT are not supported.

1) Artifact Readiness:

2) Coverage:

3) Condition-Level Metrics: Reachability is total_received_packets / total_injected_packets within the condition. Latency is received-packet-weighted across runs with received packets. Mean network throughput is the exported finite-window mean across all five runs. Status codes are: C = complete_injection_cell, P = partial_injection_cell, and E = empty_injection_cell. Blank reachability means no injected-packet denominator exists. Blank latency means no received-packet delay samples exist.

Table IV is a finite-window synthetic-traffic table. Its partial DeFT delivery ratios do not contradict the drain-mode reachability audit, because Table IV counts packets delivered before a fixed simulation cutoff under ongoing synthetic injection, whereas the reachability audit stops injection and then waits for the network to drain.

4) Pairwise Readiness: No delta columns are included. Table V describes whether values can be displayed side by side, not whether either routing mode should be preferred. The readiness code XY-empty means the XY side injected zero packets in the measured cell. The readiness code latency-blank means both modes have injection, but one mode has zero received packets, so latency comparison is blank.

5) Zero-Injection Runs: The full 54-run list is preserved in external/noxim/other/generated/t0027_report_support_v1/zero_injection_runs.csv. Table VI summarizes the zero-injection rows by routing and traffic profile.

6) Drain-Mode Post-Fix Validation and IA-XY Comparison: The drain-mode evaluations use opt-in source-cutoff plus drain semantics rather than the fixed-window policy used for Table IV. These results do not update the fixed-window condition metrics and do not relabel IA-XY as standard XY.

Table VII supports a stronger DeFT reachability statement within a precise boundary: deterministic hardcoded fixtures, seed 0, opt-in drain mode, the current physical bidirectional VL fault model, and the five accepted masks listed in the table. Within that boundary, every accepted mask delivered all 4032 valid ordered chiplet-router source-destination pairs. This is an eventual-delivery result after injection cutoff and drain completion, not a finite-window throughput or latency result.

The denominator-safe T0053 comparison table classifies 68 matched fixture/fault rows as complete_delivery_both_modes and 27 rows as descriptive_only_timeout_or_non100. The latter rows are reported only as timeout/non-100% evidence; they are not used for ranking, improvement percentages, or statistical conclusions. For the IA-XY all-pairs aggregate rows, the five fault-mask cases stopped at drain_timeout after receiving 557, 1211, 888, 995, and 795 measured packets. These rows also retained source queues and in-network state at the timeout, so they remain descriptive-only.

B. Comparative Analysis

The fixed-window comparison with standard XY is not comparison-ready for latency or performance ranking. Table V records why: hotspot rows have no measured-window XY injection, and the uniform/localized rows that do inject on the XY side have zero received packets. These rows are useful for documenting the limitation of cardinal-only XY on DEFT_2_5D, but not for computing deltas.

The later drain-mode comparison uses INTERPOSER_AWARE_XY, not standard XY, because IA-XY can traverse the modeled interposer path. Table VIII shows that DEFT passed all 95 matched rows while IA-XY passed 68 rows and timed out in 27 rows. This supports only the matrix-scoped statement that DEFT completed the documented comparison matrix and that IA-XY had descriptive timeout rows in that same matrix. It does not support universal algorithm ranking, improvement percentages, or statistical conclusions.

C. Discussion

The implementation produced a complete Noxim extension surface, a validated fixed-window synthetic sweep, and drain-

TABLE IV
CONDITION-LEVEL DESCRIPTIVE METRICS

Routing	Traffic	Mask	Fault	Nonempty	Empty	Inj.	Recv.	Reach.	Latency	Thput.	Status
DEFT	hotspot_3x10	0x0000	0	5	0	69	19	0.27536232	12.73684211	0.0034222222	C
DEFT	hotspot_3x10	0x0001	6.25	4	1	19	4	0.21052632	16.75	0.0007111111	P
DEFT	hotspot_3x10	0x0011	12.5	3	2	114	72	0.63157895	40.36111111	0.0126888889	P
DEFT	hotspot_3x10	0x0111	18.75	5	0	66	18	0.27272727	7.61111111	0.0034666667	C
DEFT	hotspot_3x10	0x1111	25	5	0	68	21	0.30882353	9.47619048	0.0043333333	C
DEFT	localized_40	0x0000	0	5	0	289	166	0.57439446	23.37951807	0.0294888889	C
DEFT	localized_40	0x0001	6.25	5	0	144	72	0.5	16.83333333	0.0129777778	C
DEFT	localized_40	0x0011	12.5	5	0	114	60	0.52631579	16.08333333	0.0112	C
DEFT	localized_40	0x0111	18.75	5	0	303	155	0.51155116	18.50967742	0.0285111111	C
DEFT	localized_40	0x1111	25	5	0	166	89	0.53614458	18.43820225	0.0160888889	C
DEFT	uniform	0x0000	0	5	0	82	38	0.46341463	24.23684211	0.0071555556	C
DEFT	uniform	0x0001	6.25	5	0	137	84	0.61313869	23.96428571	0.0149777778	C
DEFT	uniform	0x0011	12.5	5	0	63	17	0.26984127	22.23529412	0.0032888889	C
DEFT	uniform	0x0111	18.75	5	0	81	49	0.60493827	20.08163265	0.0086	C
DEFT	uniform	0x1111	25	4	1	73	50	0.68493151	21.9	0.0087555556	P
XY	hotspot_3x10	0x0000	0	0	5	0	0			0	E
XY	hotspot_3x10	0x0001	6.25	0	5	0	0			0	E
XY	hotspot_3x10	0x0011	12.5	0	5	0	0			0	E
XY	hotspot_3x10	0x0111	18.75	0	5	0	0			0	E
XY	hotspot_3x10	0x1111	25	0	5	0	0			0	E
XY	localized_40	0x0000	0	4	1	7	0	0		0	P
XY	localized_40	0x0001	6.25	4	1	7	0	0		0	P
XY	localized_40	0x0011	12.5	4	1	7	0	0		0	P
XY	localized_40	0x0111	18.75	4	1	7	0	0		0	P
XY	localized_40	0x1111	25	4	1	7	0	0		0	P
XY	uniform	0x0000	0	1	4	1	0	0		0	P
XY	uniform	0x0001	6.25	1	4	1	0	0		0	P
XY	uniform	0x0011	12.5	1	4	1	0	0		0	P
XY	uniform	0x0111	18.75	1	4	1	0	0		0	P
XY	uniform	0x1111	25	1	4	1	0	0		0	P

TABLE V
XY/DEFT PAIRWISE READINESS

Traffic	Fault mask	XY status	DEFT status	Readiness
hotspot_3x10	0x0000	E	C	XY-empty
hotspot_3x10	0x0001	E	P	XY-empty
hotspot_3x10	0x0011	E	P	XY-empty
hotspot_3x10	0x0111	E	C	XY-empty
hotspot_3x10	0x1111	E	C	XY-empty
localized_40	0x0000	P	C	latency-blank
localized_40	0x0001	P	C	latency-blank
localized_40	0x0011	P	C	latency-blank
localized_40	0x0111	P	C	latency-blank
localized_40	0x1111	P	C	latency-blank
uniform	0x0000	P	C	latency-blank
uniform	0x0001	P	C	latency-blank
uniform	0x0011	P	C	latency-blank
uniform	0x0111	P	C	latency-blank
uniform	0x1111	P	P	latency-blank

TABLE VI
ZERO-INJECTION RUN SUMMARY

Routing	Traffic	Zero-injection runs
DEFT	hotspot_3x10	3
DEFT	uniform	1
XY	hotspot_3x10	25
XY	localized_40	5
XY	uniform	20

mode reachability evidence for DeFT. The principal interpretation constraint is the denominator of each measurement regime. In the fixed-window sweep, aggregated reachability is undefined when no packets were injected, and latency is undefined when no packets were received. Empty or partial injection cells therefore support descriptive reporting but not unqualified performance claims.

The stronger DeFT reachability claim is limited to the deterministic drain-mode audit in Table VII. It does not extend to stochastic traffic, finite-window throughput, universal timing, PARSEC/GEM5 workloads, single-direction endpoint fault modeling, or IA-XY behavior. The fixed-window Table IV and the drain-mode audit answer different questions, so their results should be read as complementary rather than contradictory.

The comparison baseline also has a scope constraint. Standard XY is cardinal-only on DEFT_2_5D and does not traverse the VL/hub/interposer paths required for unrestricted inter-chiplet traffic. The drain-mode comparison therefore uses INTERPOSER_AWARE_XY, which must remain distinct from

TABLE VII
DEFT DRAIN-MODE ALL-PAIR REACHABILITY AUDIT

Item	Value
Routing mode	DEFT
Drain policy	Opt-in source cutoff plus drain
Traffic fixtures	130 deterministic hardcoded fixtures
Seed	0
Fault model	Current physical bidirectional VL model
Accepted masks	0x0000, 0x0001, 0x0011, 0x0111, 0x1111
Simulator cases	650
Pass cases	650
Timeout or non-100% cases	0
Valid ordered pairs per mask	4032 / 4032 delivered
Accepted masks with full pair delivery	5 / 5
Source fix indicated	No

TABLE VIII
T0053 IA-XY DRAIN-MODE OUTCOME SUMMARY

Fixture group	Rows	IA-XY pass	IA-XY timeout
route_family_pair	40	40	0
source_isolated	20	20	0
destination_stress	20	6	14
bounded_aggregate	5	1	4
bounded_low_load	5	1	4
all_pairs_aggregate	5	0	5
Total	**95**	**68**	**27**

standard XY. Universal rankings, improvement percentages, latency deltas, and statistical conclusions require a broader validation policy than the current artifacts provide.

D. Reproducibility

The final LaTeX artifact is generated from final_report/main.tex, which was derived from docs/FINAL_REPORT_DRAFT.md and uses the same IEEE conference-style report template family as the extended proposal. The final report source is intentionally separated from the extended proposal source so the original archive and proposal files remain unchanged.

The code and experiment configuration are in the project repository, with the registered Noxim source tree at external/noxim. The validated Noxim build command is ./build.sh from external/noxim in the documented WSL/Linux environment. The final fixed-window experiment commands, generated LUTs, JSON statistics, manifests, and summaries are preserved under external/noxim/other/generated/t0026_final_sweep_v1/, external/noxim/other/generated/t0026_final_analysis_v1/, external/noxim/other/generated/t0027_report_support_v1/, and external/noxim/other/generated/t0028_final_report_results_v1/. The drain-mode reachability and comparison artifacts are preserved under external/noxim/other/generated/t0060_deft_reachability_audit_v1/ and external/noxim/other/generated/t0053_drain_iaxy_deft_comparison_v1/; each artifact set includes the relevant commands, fixtures, generated LUTs, logs, summaries, and manifests.

The final PDF was generated from final_report/ using the documented TeX fallback sequence pdflatex main.tex, bibtex main, pdflatex main.tex, and pdflatex main.tex. The simulator final sweep was not rerun during final-report conversion, report revision, report integration, or packaging refresh.

IX. CONCLUSION AND FUTURE WORK

The project produced a traceable Noxim extension for DEFT_2_5D and a reproducible evaluation artifact set. The fixed-window synthetic sweep confirms that the planned 150-run matrix executed successfully and that generated summary tables cross-check against raw artifacts. Those fixed-window results remain descriptive because the measured data contains blank and partial cells and because standard XY is not an interposer-aware unrestricted baseline for this topology.

The strongest DeFT correctness evidence comes from the opt-in drain-mode all-pair audit: within deterministic hardcoded fixtures, seed 0, the current physical bidirectional VL fault model, and the accepted fault-mask ladder, all 650 DEFT cases passed and all 4032 valid ordered source-destination pairs delivered for every accepted mask. No source fix is indicated by that audit. Future work should address IA-XY timeout diagnosis, broader stochastic and statistical evaluation, directional endpoint fault modeling, traffic-profile-specific LUT demand inputs, non-empty hotspot measurements for standard XY control runs, and validated PARSEC/GEM5 trace support before expanding latency, throughput, ranking, or workload-level claims.

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